EliteDesign: Elite-Guided Learning for RNA Design via the Cross-Entropy Method

JasonFred Ngwa Milan Gautam David H. Mathews Liang Huang

Abstract

The goal of RNA design is to find, within an exponentially large sequence space, a nucleotide sequence that folds into a specified target structure. While essential for developing functional RNAs such as riboswitches and ribozymes, efficiently identifying sequences with high target folding probability remains challenging, especially for long sequences. Currently, the state-of-the-art method, SamplingDesign, is computationally very expensive. To address this bottleneck, we present EliteDesign, a novel framework that leverages the cross-entropy method to iteratively refine a distribution over candidate RNA sequences. On the Eterna100 benchmark, EliteDesign achieves a higher average target folding probability than SamplingDesign while running 180× faster. Furthermore, EliteDesign scales efficiently to long RNA structures far beyond the reach of SamplingDesign: on ten 16S rRNA targets (950–1,525 nucleotides), it achieves an average target folding probability of 0.52 versus 0.20 for the strongest scalable baseline, SAMFEO, while accelerating runtime by 14×.

1 Introduction

RNA plays two main roles in cells. It can serve as an information carrier, carrying the genetic code needed to make proteins, but it can also serve a functional role such as catalyzing chemical reactions or regulating genes (Breaker, 2012; Doudna and Cech, 2002). Recent advances have shown that RNA has huge potential for developing new medicines. For example, Rojas et al. (2023) showed that personalized mRNA vaccines stimulated immune responses against tumor targets in patients with pancreatic cancer. This advancement could open the path toward cancer treatments that are tailored to individual patients. Additionally, on the computational side, recent methods such as LinearDesign have shown that algorithmic techniques can be used to design mRNAs with improved stability, protein production, and immune response (Zhang et al., 2023). These advances in RNA technology motivate further work on designing RNA.

RNA design, also called inverse folding, is the task of finding a nucleotide sequence that folds into a given target structure (Dirks et al., 2004; Hofacker et al., 1994). Given a target structure, a good RNA design method should find a sequence that favors the target over the competing folds in its ensemble. However, this problem is difficult because the number of possible sequences for a target grows exponentially with length, and each of these sequences can adopt many alternative structures, which makes the search especially hard for long RNAs (Tang et al., 2026). Existing methods that optimize target probability face limitations in quality, runtime, or both. For example, SAMFEO maintains several promising sequences and uses structure-aware mutations to optimize folding probability (Zhou et al., 2023), but it struggles on difficult and long targets, where the enormous design space makes local search prone to getting stuck in local optima. On the other hand, SamplingDesign (Tang et al., 2026) produces high-quality designs, even for difficult puzzles in Eterna100, but it is computationally expensive and too slow to be practical for longer RNAs. Finding a high-probability sequence within a practical runtime remains a major challenge.

To address this bottleneck, we present EliteDesign, which uses the cross-entropy method (de Boer et al., 2005; Rubinstein, 1999) to find RNA sequences with high target probability in a computationally efficient way. At each iteration, EliteDesign samples from the valid design space, scores the sequences, and then picks a set of elite sequences based on target probability. It then moves the current distribution toward the distribution described by the elite set. During the sampling rounds, sequences are scored with LinearPartition (Zhang et al., 2020), which estimates folding probability in linear time, and the best final candidates are rescored exactly with ViennaRNA (Lorenz et al., 2011).

We benchmark EliteDesign on several datasets and show that it finds higher target folding probabilities than the methods we compare against, with the gap widening on long RNA sequences. These results show that EliteDesign improves both design quality and runtime compared with methods that optimize target folding probability, and allows us to find high-probability sequences for much longer RNA structures.

2 Methods

2.1 RNA design problem

2.1.1 Sequences, structures, and valid designs

To approach the RNA design problem, we first define the valid design space for a target structure. A sequence of length n can be represented as $x = x_1 \cdots x_n$, where each nucleotide x_i is chosen from $\mathcal{A} = \{\mathrm{A}, \mathrm{C}, \mathrm{G}, \mathrm{U}\}$. Given a target structure y, we can further restrict the base choices using two sets: $\mathcal{P}(y)$ for paired positions and $\mathcal{U}(y)$ for unpaired positions (Tang et al., 2026). Each pair in $\mathcal{P}(y)$ must be one of the canonical or wobble pairs:

$$\mathcal{C} = \{\mathrm{AU}, \mathrm{UA}, \mathrm{CG}, \mathrm{GC}, \mathrm{GU}, \mathrm{UG}\}. \tag{1}$$

The valid design space can then be represented as

$$\begin{aligned}\mathcal{X}(y) = \{\, x \in \mathcal{A}^n :\ & x_i x_j \in \mathcal{C} \\ & \text{for every } (i,j) \in \mathcal{P}(y)\}.\end{aligned} \tag{2}$$

This restriction greatly reduces the design space for structures with many pairs by removing incompatible choices, such as C pairing with A. The number of sequences in the design space for a target y with p pairs and u unpaired positions is then

$$|\mathcal{X}(y)| = 6^p 4^u, \qquad |y| = 2p + u. \tag{3}$$

2.1.2 Target folding probability

Now that we have a valid design space for the target y, we need a way to assess how well a given sequence favors folding into the target. To do this, we calculate the target folding probability, which is the probability that the sequence folds into the target structure, given its ensemble of competing structures. Let $G(x, y)$ be the free energy of a sequence x folded into structure y, which is calculated using the Turner 2004 nearest-neighbor energy model (Mathews et al., 2004). The lower the free energy, the more favorable the structure is for that sequence. The folding ensemble for sequence x is

$$\mathcal{Y}(x) = \{y' : y' \text{ is an allowed structure for } x\}. \tag{4}$$

This set contains the target structure and all competing folds that the sequence can fold into. The total statistical weight of all structures for the sequence is given by the partition function (McCaskill, 1990),

$$Z(x) = \sum_{y' \in \mathcal{Y}(x)} e^{-G(x,y')/(RT)}. \tag{5}$$

Here R is the gas constant and T is the temperature; the product RT is usually expressed in kcal/mol. Based on this formula, the lower the energy of a structure, the more it contributes to the ensemble weight.

Target folding probability can then be defined as

$$P(y \mid x) = \frac{e^{-G(x,y)/(RT)}}{Z(x)}. \tag{6}$$

This equation divides the target's Boltzmann weight by the total weight of all allowed structures, giving the probability that the sequence forms the target under the thermodynamic model.

The primary design objective of EliteDesign is

$$x^\star \in \arg\max_{x \in \mathcal{X}(y)} P(y \mid x). \tag{7}$$

During search we rank candidates by log probability,

$$\ell(x) = \log P(y \mid x) = -\frac{G(x,y)}{RT} - \log Z(x), \tag{8}$$

We optimize in log space to avoid underflow when probabilities are very small.

2.1.3 Normalized ensemble defect

Normalized ensemble defect (NED) is the expected fraction of nucleotides whose pairing state differs from the target structure y, averaged over all folds y' in the ensemble (Dirks et al., 2004; Zadeh et al., 2011). For positions i and j, the pairing probability is

$$p_{ij}(x) = \sum_{(i,j) \in \mathcal{P}(y')} P(y' \mid x). \tag{9}$$

The probability that position i is unpaired is

$$u_i(x) = 1 - \sum_{j \neq i} p_{ij}(x). \tag{10}$$

Let $\mathcal{P}$ and $\mathcal{U}$ be the sets containing the target's paired and unpaired positions respectively. Then

$$\mathrm{NED}(x) = 1 - \frac{1}{n}\left[2\textstyle\sum_{(i,j)\in\mathcal{P}} p_{ij}(x) + \sum_{i \in \mathcal{U}} u_i(x)\right]. \tag{11}$$

Each correct pair is counted twice because it contains two nucleotides.

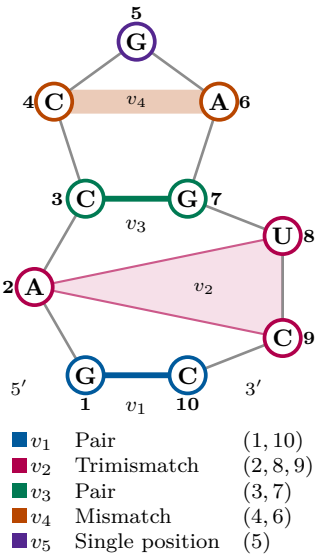


Figure 1: Example of a target structure being split into variables.

2.2 Structure parsing and distribution initialization

Following SamplingDesign (Tang et al., 2026), given a target structure in dot-bracket notation, for example

```
( . ( . . . ) . . )
1 2 3 4 5 6 7 8 9 10
```

we parse the structure into four types of variables:

- **Unpaired variables** with four nucleotide choices: A, C, G, or U. For example, position 5 is an unpaired variable.
- **Paired variables** sampled from AU, UA, CG, GC, GU, or UG. For example, positions $(1, 10)$ form a paired variable.
- **Mismatch variables** group the two unpaired positions next to the pair that closes a loop, and have $4^2 = 16$ choices. For example, positions $(4, 6)$ form a mismatch variable.
- **Trimismatch variables** group three unpaired positions in a loop, one on one side and two on the other side. These bases are chosen together, giving $4^3 = 64$ possible combinations. For example, positions $(2, 8, 9)$ form a trimismatch variable.

To assign initial probabilities to these variables, we follow the targeted initialization used in SamplingDesign (Tang et al., 2026). This initialization favors A at unpaired positions and CG or GC at paired positions, with the parameter $\epsilon \in [0, 1]$ controlling the strength of this preference. For an unpaired variable, the initial probabilities are

$$p(b) = \begin{cases} \epsilon + (1-\epsilon)/4, & b = \mathrm{A}, \\ (1-\epsilon)/4, & b \in \{\mathrm{C}, \mathrm{G}, \mathrm{U}\}. \end{cases} \tag{12}$$

Table 1: Unpaired variable probabilities for $\epsilon = 0.75$.

Nucleotide	Probability
A	0.8125
C	0.0625
G	0.0625
U	0.0625

For a paired variable, the initial probabilities are

$$p(s) = \begin{cases} (1+2\epsilon)/6, & s \in \{\mathrm{CG}, \mathrm{GC}\}, \\ (1-\epsilon)/6, & \text{otherwise, } s \in \mathcal{C}. \end{cases} \tag{13}$$

Table 2: Paired variable probabilities for $\epsilon = 0.75$.

Base pair	Probability
CG	0.4167
GC	0.4167
AU	0.0417
UA	0.0417
GU	0.0417
UG	0.0417

Mismatch and trimismatch variables are initialized by multiplying the unpaired probabilities of their bases.

2.3 Sampling, scoring, and elite weighting

At each round, EliteDesign samples N sequences from the current distribution with a small amount of uniform mixing, so that every variable choice remains possible:

$$p_{\text{sampling}} = (1-\delta)p_{\text{current}} + \delta U. \tag{14}$$

In this equation, U is the uniform distribution for a variable and δ is the uniform mixing rate. Each sequence in the sample is scored with LinearPartition (Zhang et al., 2020) and cached to avoid scoring the same sequence twice. The best-scoring sequence is also tracked separately and, at the end of the search, is rescored with ViennaRNA for accuracy (Lorenz et al., 2011).

Sequences in a sample are sorted by target probability and the top fraction ρ are selected as elites:

$$N_{\text{elite}} = \lceil \rho N \rceil. \tag{15}$$

Each elite is then given a weight based on its NED:

$$w_{\text{elite}}(x) = \frac{1 - \text{NED}(x)}{\sum_{j=1}^{N_{\text{elite}}} (1 - \text{NED}(x_j))}. \tag{16}$$

2.4 Distribution updates and stopping criteria

From the weighted elites we get the distribution p_{elites}, which we combine with the current distribution p_{current} using the smoothed cross-entropy update (de Boer et al., 2005):

$$p_{\text{updated}} = (1 - \alpha) p_{\text{current}} + \alpha p_{\text{elites}}. \tag{17}$$

Here α is the elite update rate, which is set to 0.5 by default in our implementation. At each round we track the sequence with the highest probability seen so far:

$$P_{\text{best}} = \max_{x \text{ sampled so far}} P(y \mid x). \tag{18}$$

We also track the moving average over h rounds:

$$P_{\text{avg}} = \frac{1}{h} \sum_{r=1}^{h} P_r, \tag{19}$$

where P_r is the average target probability in round r. We stop when both of these conditions hold:

1. The best probability has not improved for h rounds.
2. The moving average has not improved for h rounds.

Table 3: Default settings of EliteDesign.

Setting	Symbol	Default
Samples per round	N	500
Elite fraction	ρ	0.1
Elite update rate	α	0.5
Uniform mixing	δ	0.002
Initialization bias	ϵ	0.75
Stopping condition	h	25

2.5 Algorithm summary

Algorithm 1 summarizes EliteDesign.

Algorithm 1 Elite-guided RNA design

```
Input: target y, initial distribution
x* ← None
for r = 1, ..., R_max do
    Sample N sequences
    Score the sequences
    Update x* if any sequence has higher P(y | x)
    Select N_elite sequences
    Weigh elites
    Update p_current
    if stopping criterion is met then
        break
    end if
end for
Validate x* with ViennaRNA
return x*
```

3 Results

3.1 Eterna100

We evaluate EliteDesign on Eterna100, a benchmark of 100 puzzles from the Eterna game, assembled by Anderson-Lee and colleagues (Anderson-Lee et al., 2016). It contains mostly synthetic RNA secondary structures that range from 12 to 400 nucleotides and vary in design difficulty, allowing us to compare methods across diverse targets. We compare EliteDesign with SamplingDesign (Tang et al., 2026), SAMFEO (Zhou et al., 2023), and NEMO (Portela, 2018) (Table 4). We run EliteDesign with these settings:

1. Population sizes of 100 and 500, with uniform mixing of 0.2% and elite update rate $\alpha = 0.5$.
2. Five random seeds per target on a 96-core CPU; for each metric, we select the best seed.
3. Runtime is the mean time per target across the five seeds.
4. We use LinearPartition (Zhang et al., 2020) for scoring during search and ViennaRNA (Lorenz et al., 2011) for final scoring.

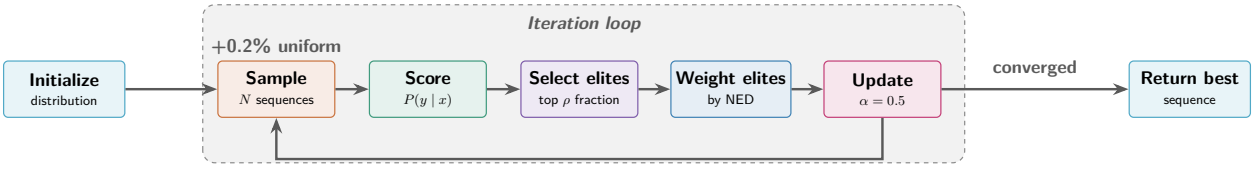


Figure 2: Overview of the EliteDesign algorithm.

Method	Mean $p(y \mid x)$ ↑	Geom. mean $p(y \mid x)$ ↑	NED ↓	Structural distance ↓	Mean energy gap $\Delta\Delta G^\circ$ ↓	MFE/ uMFE ↑	Time (s) ↓
EliteDesign100	0.6032	0.5601	0.03145	1.62	0.525	78/74	**4.21**
EliteDesign500	**0.6105**	**0.5825**	**0.03015**	**1.50**	**0.5024**	**81/77**	11.99
SamplingDesign	0.5950	0.5225	0.03518	2.15	0.72	80/76	2160
SAMFEO	0.581	0.167	0.043	4.57	2.48	77/74	279.1
NEMO	0.271	0.083	0.098	1.93	1.31	79/**77**	—

Table 4:

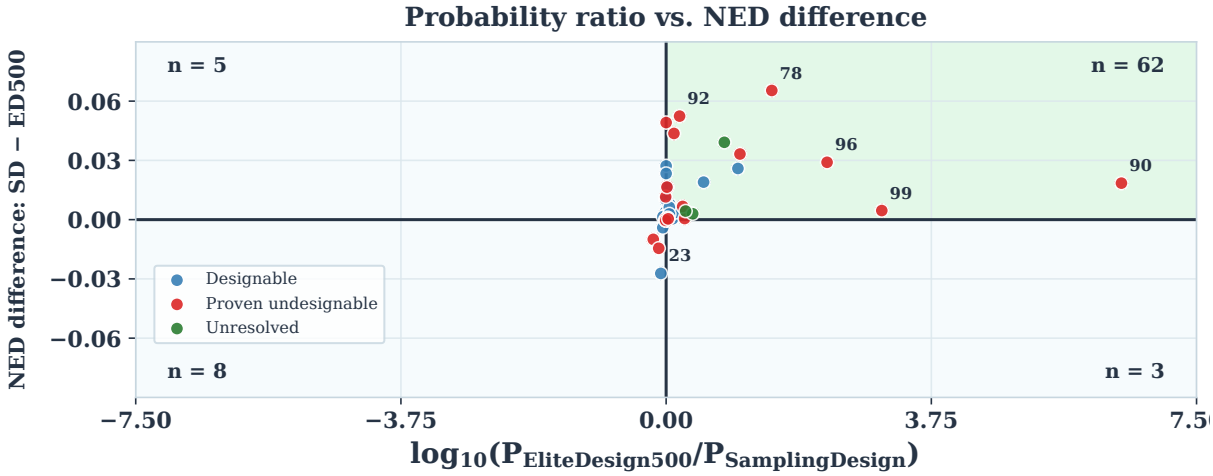


Figure 3:

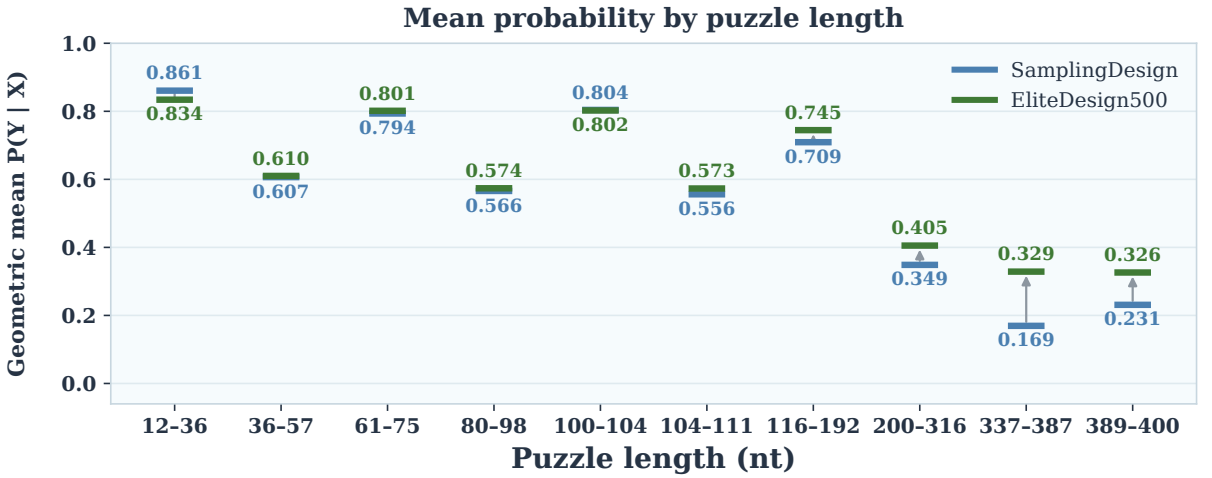
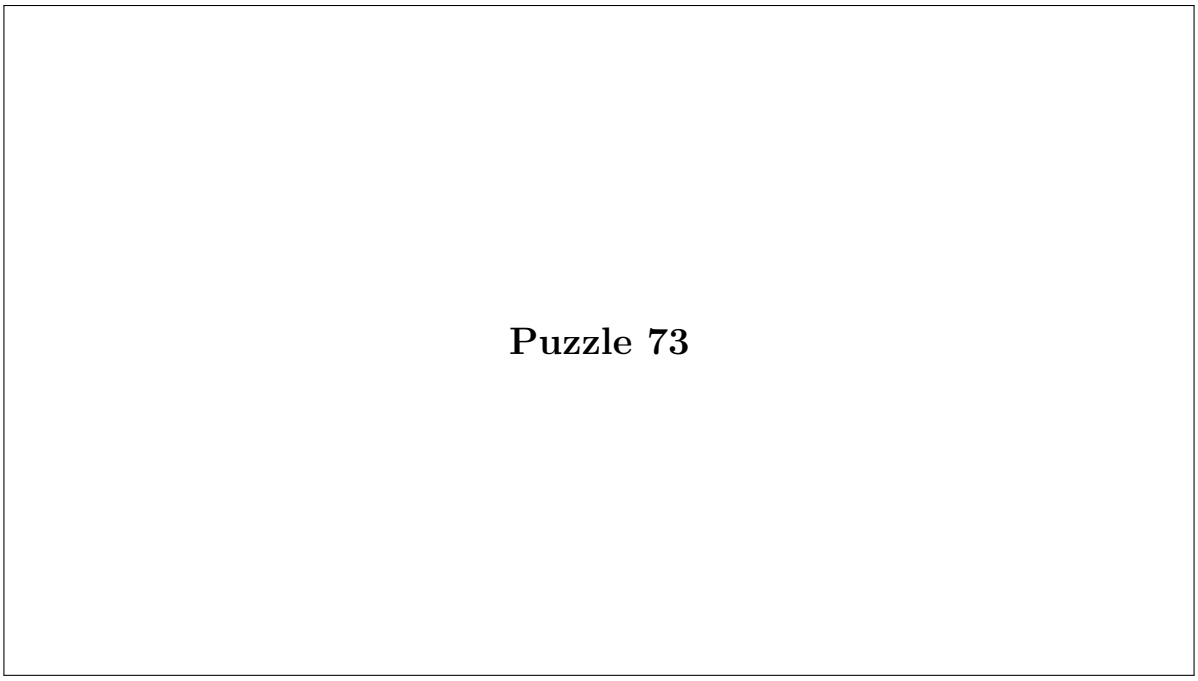
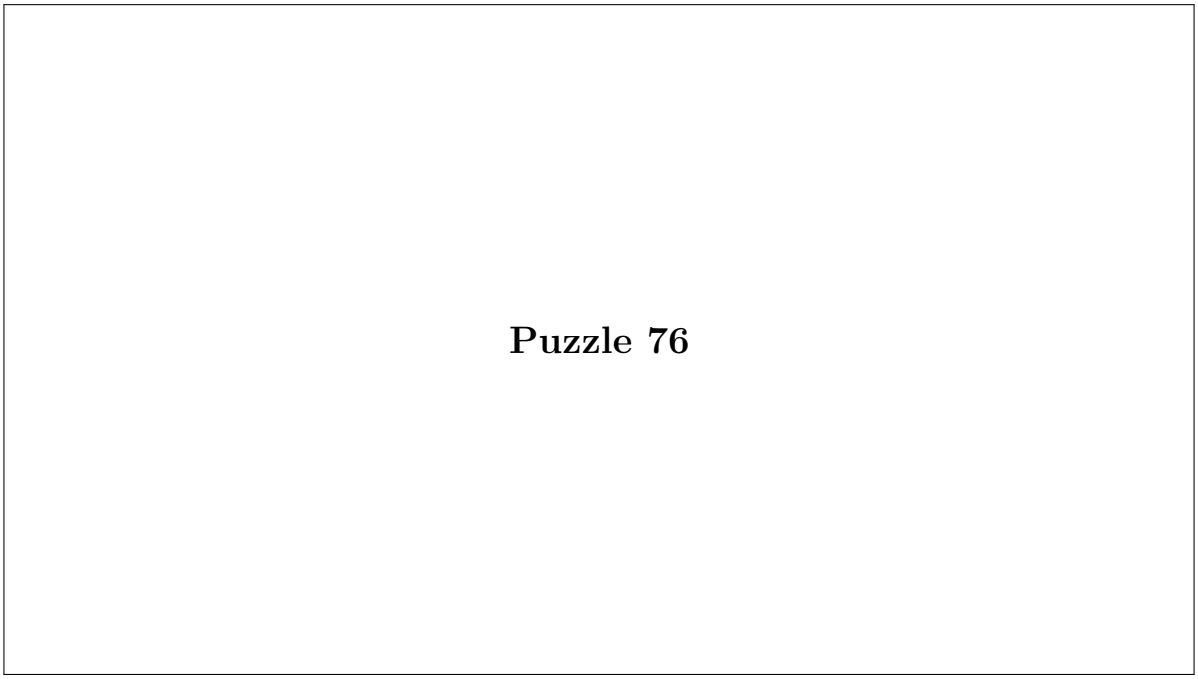


Figure 4:



(a)



(b)

Figure 5:

3.2 16S rRNA

		Target probability $p(y \mid x) \uparrow$			NED ↓			Time (min) ↓		
Target	**nt**	SAMFEO	EliteDesign 100	EliteDesign 500	SAMFEO	EliteDesign 100	EliteDesign 500	SAMFEO	EliteDesign 100	EliteDesign 500
1	950	0.290	**0.575**	0.570	0.0026	0.0013	**0.0012**	287	**9.8**	17.1
2	954	0.511	0.773	**0.779**	0.0018	**0.0006**	**0.0006**	285	**8.6**	22.8
3	1200	0.229	0.587	**0.593**	0.0028	0.0010	**0.0009**	603	**8.8**	35.0
4	1452	0.010	0.192	**0.200**	0.0079	0.0028	**0.0027**	1,091	**48.5**	113
5	1474	0.086	0.344	**0.377**	0.0040	0.0016	**0.0014**	1,150	**43.1**	51.4
6	1474	0.247	0.553	**0.588**	0.0024	0.0010	**0.0009**	1,263	**48.6**	59.2
7	1490	0.152	0.540	**0.542**	0.0038	**0.0010**	**0.0010**	1,165	**30.1**	55.1
8	1492	0.276	0.652	**0.654**	0.0023	**0.0007**	**0.0007**	1,216	**68.0**	105
9	1497	0.143	0.406	**0.410**	0.0038	**0.0017**	**0.0017**	1,199	**62.0**	108
10	1525	0.071	0.474	**0.484**	0.0044	**0.0012**	**0.0012**	1,198	**67.0**	93.0
Mean		0.202	0.510	**0.520**	0.0036	0.0013	**0.0012**	946	**39.4**	66.0

Table 5: 16S rRNA results.

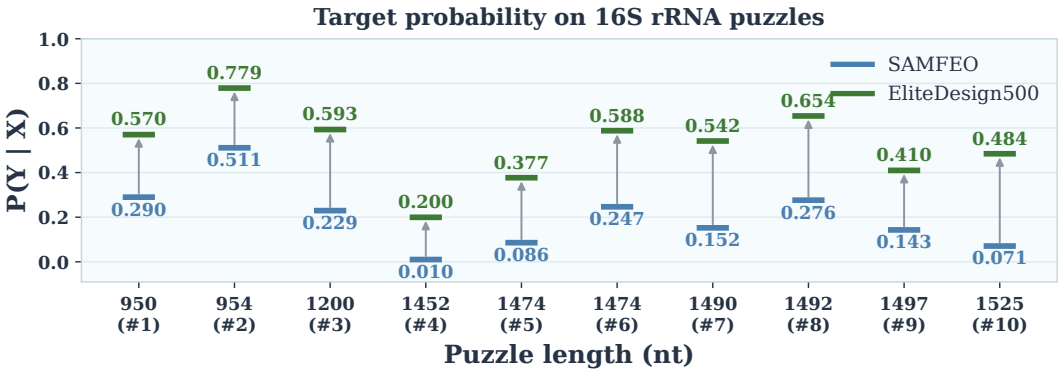


Figure 6:

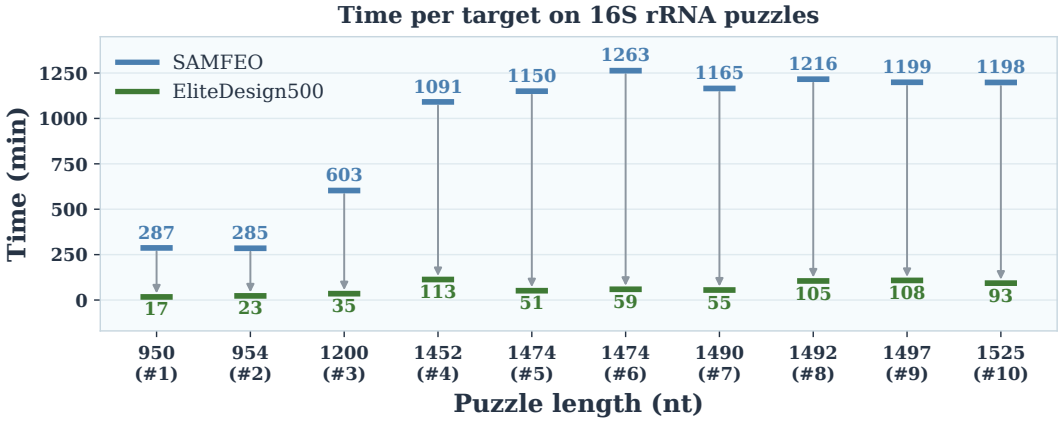


Figure 7:

4 Discussion

5 Conclusion

In conclusion, this work introduces EliteDesign, an RNA design framework that uses the cross-entropy method to find high-probability RNA sequences for a given target structure. We further show that this method addresses a gap in current RNA design methods by enabling the design of sequences longer than 1,000 nucleotides with high target folding probability. Our method beats the state-of-the-art SamplingDesign on Eterna100 in both speed and quality. Additionally, we show large gains over SAMFEO on ten 16S rRNA targets while remaining much faster per puzzle. We also show that our method achieves higher target probability on other RNA design datasets such as Rfam (Kalvari et al., 2021) and RNAsolo764 (Adamczyk et al., 2022). Despite these gains, EliteDesign has some limitations:

1. EliteDesign uses the same initialization strategy for every target, which may not work equally well for all structures. During optimization, it can concentrate too early on a narrow set of sequences and miss better solutions.

2. EliteDesign uses LinearPartition to estimate folding probabilities during search. Errors in these estimates can cause it to favor weaker candidates and miss better ones. Exact scoring at the end checks the final candidates but cannot recover sequences missed during search.

Future work could add a local refinement stage similar to SAMFEO (Zhou et al., 2023), which might further improve NED and target probability on some structures.

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